

Beyond Character Matching: Proposing Psycholinguistically Informed Metrics for Chinese Radical Knowledge in LLMs

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Abstract

Chinese characters exhibit rich internal structure: basic graphemes (radicals) combine compositionally to form complex characters, encoding semantic, phonetic, and structural information. While this sub-character structure is central to human literacy and to models’ ability to generalize, most contemporary Chinese language models treat characters as atomic symbols. Moreover, existing evaluations rely primarily on downstream tasks dominated by common characters, leaving it unclear whether models can generalize sub-character knowledge to domain-specific settings involving rare characters.

We introduce a psycholinguistically motivated framework for probing radical awareness in Chinese language models. Our approach involves training a gold standard radical-aware model, a radical awareness metric of a given model, and controlled character inference tasks with human judgments. As a pilot study, we apply these tools to contemporary Chinese LMs, providing initial evidence on the extent to which radical-based compositionality is reflected in their representations and outlining a path toward more diagnostic evaluation benchmarks.

1 Introduction

Chinese characters are inherently compositional. They exhibit a hierarchical internal structure in which basic graphemes—commonly referred to as **radicals**—combine to form sub-characters and full characters. Radicals often encode semantic, phonetic, or structural information and recur systematically across the writing system. This rich sub-character structure plays a central role in human literacy and language processing.

Despite this, most contemporary natural language processing systems treat Chinese characters

as atomic symbols, analogous to letters in alphabetic scripts. By ignoring sub-character structure, these models fail to exploit a key source of linguistic information in written Chinese. This limitation becomes particularly salient in settings where rare characters occur, and surface-level memorization is insufficient.

Indeed, while large language models (LLMs) demonstrate impressive fluency in modern Chinese, they consistently underperform on domain-specific tasks such as classical Chinese understanding or on out-of-domain vocabulary, including rare words and toponyms (Wu et al., 2025). These failures suggest that current models lack a principled mechanism for generalizing beyond observed character forms. Existing benchmarks for evaluating Chinese language models primarily rely on downstream tasks such as cloze tests or translation. Such evaluations are coarse-grained and general: they conflate multiple linguistic competencies and mostly use common characters. As a result, they are ill-suited for diagnosing weaknesses related to sub-character processing.

We argue that many of these observed deficiencies are closely tied to a model’s ability to access and exploit sub-character information—an ability we term **radical awareness**. To address this gap, we propose a set of psycholinguistically motivated evaluation metrics that directly probe radical awareness. These metrics aim to provide a more faithful and fine-grained assessment of a model’s knowledge of Chinese characters, complementing existing task-based benchmarks.

2 Related work

Compositionality in Chinese characters

Sinitic languages are natively written using Chinese characters, known as Hànzì. Each character is composed of subunits, typically groups of strokes that recur across many characters and often serve semantic, phonetic, or structural func-

tions. In this work, we refer to these graphemes as **radicals**. The compositional analysis of Hànzì dates back to classical lexicographic works such as the *Zihui* (Mei, 1615) and later developments including the *Kāngxī Dictionary* (Zhang and Chen, 1716) and early modern analyses (Wieger, 1915). Modern resources include Myers (2019) that discussed the relevance of character decomposition supported by an overview of empirical evidence.

Psycholinguistics evidence for radical information in Hanzì Various psycholinguistics have shown that native Chinese speakers exhibit radical awareness from an early age. Even young learners can identify and manipulate character constituents without necessarily knowing the characters’ meanings or pronunciations (Liu et al., 2010; Anderson et al., 2013; Shu and Anderson, 1999). This sensitivity to sub-character structure plays a crucial role in reading acquisition, writing, and vocabulary learning (Liu, 2013; McBride-Chang et al., 2003, 2008; Tsai and Nunes, 2003), and the decomposition happens in real time (Hsiao and Cottrell, 2009). These findings underscore the centrality of radical-level processing in human cognition and motivate its explicit consideration when evaluating computational models of Chinese.

Existing benchmarks and evaluation metrics for Chinese Various benchmarks have been introduced to evaluate the Chinese language capabilities of large language models; however, most remain task-centric and operate at the word or sentence level. CLUE (Chinese Language Understanding Evaluation) is an early and influential benchmark suite covering core natural language understanding tasks such as text classification, semantic similarity, natural language inference, and reading comprehension (Xu et al., 2020). CUGE (Chinese Understanding and Generation Evaluation) extends this scope to both comprehension and generation, including summarization, dialogue generation, question generation, and machine reading comprehension (Yao et al., 2022). More recently, C-Eval assesses advanced knowledge and reasoning through multi-disciplinary, exam-style multiple-choice questions in Chinese (Huang et al., 2023), while CIF-Bench focuses on instruction following and open-ended generation, emphasizing robustness and generalization across diverse tasks (Li et al., 2024). These benchmarks, despite their

large scale, do not consider sub-character information.

Wu et al. (2025) investigates the impact of incorporating sub-character information into Chinese language models. Their evaluation includes stroke decomposition, structure, and radical recognition, as well as downstream tasks such as part-of-speech tagging and named entity recognition. Using radical-aware prompting, they show improvements in both sub-character probing tasks and downstream performance. However, these evaluations primarily focus on classical NLP tasks, while our work comes from a cognitively motivated perspective, testing *radical awareness* via simulating the human learning process.

3 Methodology

3.1 Radicalized representation and corpus construction

Building on existing decomposition dataset (Kawabata et al., 2018), we represent each character as a decomposition tree over radicals¹ (see Figure 1). We then linearize these trees into strings (e.g., bracketed or operator–argument notation), yielding a corpus whose basic vocabulary is radicals plus combinators rather than surface characters.

Starting from a large Chinese text corpus (e.g., news, web, literature), we map every character into its radical-tree string. We then define a set of *test characters* consisting of: (i) characters selected by graded proficiency levels (e.g., HSK bands) to cover a range of frequencies; and (ii) rare and archaic characters that remain compositional but are infrequent in modern text. We construct:

- a **training set**, obtained by removing all sentences that contain any test character (so the model never observes those characters in training, in any form); and
- a **test set**, consisting of sentences that contain at least one test character (plus additional material for rare/archaic characters).

We then train an autoregressive or sequence-to-sequence model purely on this “radicalized”² training corpus. Any success on the test characters must

¹Optionally, we can include structural operators, such as horizontal or vertical concatenation. The study of the amount of information they contain is ongoing (Liu et al., 2025)

²Here we use radicalized to refer to a character representation in which each Chinese character is deterministically decomposed into its constituent radicals, and these radicals (rather than whole characters) form the model’s basic input

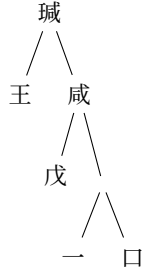


Figure 1: Tree representation of 璵

arise from composition over radicals and contextual reasoning, rather than memorization of those characters.

3.2 Radical inference tasks

We adapt lexical-proficiency evaluations from Ciccio et al. (2025) to the Chinese setting. All tasks are evaluated on held-out or rare/archaic characters.

(a) Character generation from definition.

- **Input:** A short natural-language definition describing a concept plausibly expressible by a single character or compositional pseudo-character.
- **Output:** A radical-tree string corresponding to (i) an attested but held-out character, or (ii) a formally well-formed pseudo-character under the **radical grammar**, i.e., the constraints governing which components can combine and in what configurations to form well-formed characters.

Input: a type of jade used as an ornament
Output: 璵 = 王(戊(一口))

For held-out characters, we measure exact-match and top- k accuracy against the gold character tree string. For pseudo-characters, human annotators judge radical appropriateness and semantic plausibility. This task probes whether the model can select suitable radicals and combine them into well-formed structures on the basis of definitions alone.

(b) Definition modeling in context.

and output units. Accordingly, a radicalized corpus is a corpus rewritten in this representation, and a radicalized model is a model trained to operate over it.

- **Input:** a test-set sentence containing a held-out or rare character, where the target character is given in radical form and its position in the sentence is marked.
- **Output:** a natural-language definition of the target character *as used in that sentence*.

Input: 又东二百里,曰葛山,其上多赤金,其下多璵石。
Two hundred leagues further east is Geshan Mountain. Its top is full of red gold. Its foot is full of jade stones)

Output: a type of jade
 (or, more coarsely, a kind of gemstone)

Human raters evaluate (i) coarse semantic correctness (e.g., identifying “some kind of gem” vs. a specific jade term) and (ii) contextual appropriateness (word-sense choice). This measures whether the model can infer the approximate meaning of an unseen character from its radicals and sentential context, particularly when radicals are semantically ambiguous.

(c) Usage / exemplification.

- **Input:** a target character (held-out, rare, or pseudo-character) in radical form, along with a short gloss.
- **Output:** a sentence that uses the character in a way that is semantically compatible with the gloss.

Input: 王(戊(一口))
Gloss: a kind of jade used for ornaments
Output: **ACCEPTABLE.** 他佩着一枚用璵雕成的坠饰。
He wears a pendant carved from 璵.
UNACCEPTABLE. 璵令一下, 众人立刻噤声。
Once the 璵 issues an order, everyone immediately falls silent.

Figure 2: Illustrative acceptable vs. unacceptable outputs for task (c). The unacceptable output is fluent but conflicts with the gloss by coercing the target into an authority-related predicate (misreading 王 as “king” rather than a jade classifier).

Human raters judge whether the sentence (i) is grammatical and natural and (ii) uses the target character in a way that matches the provided gloss.

Rare and archaic characters play a dual role: they stress-test generalization to genuinely unseen items, and they exploit the fact that older forms often display more transparent radical–meaning relations than their modern descendants.

3.3 Latent Radical Score(LRS)

Once validated on the above tasks, the radicalized model provides radical-aware embeddings and a “gold standard” notion of structural similarity. We then define a metric that measures the extent to which standard Chinese LMs, trained over characters rather than radicals, encode radical and orthographic structure in their embedding spaces³ beyond purely semantic similarity.

For a target character c , we construct triplets (c, p, n) subject to:

- p : **compositionally similar** to c (shares key radicals/structure under the radical grammar) and **semantically related** (near-synonym or close semantic neighbor);
- n : **semantically related** to c to a comparable degree as p , but **compositionally dissimilar** (no shared radicals or clearly different structure).

Formally, let $\mathbf{r}(\cdot)$ be the radical embedding of the radicalized model. Let $\mathbf{e}_m(\cdot)$ and $\mathbf{e}_f(\cdot)$ be the intermediate layer and final layer embedding of the target model, respectively. Let $\mathcal{C}$ be the set of characters we consider. Our choice of the triplets (c, p, n) should satisfy the following criteria.

$$\begin{aligned} \cos_sim(\mathbf{r}(c), \mathbf{r}(p)) &> \cos_sim(\mathbf{r}(c), \mathbf{r}(n)) \\ \cos_sim(\mathbf{e}_f(c), \mathbf{e}_f(n)) &\gtrsim \cos_sim(\mathbf{e}_f(c), \mathbf{e}_f(p)) \\ &\gg \frac{1}{|\mathcal{C}|} \sum_{x \in \mathcal{C}} \cos_sim(\mathbf{e}_f(c), \mathbf{e}_f(x)) \end{aligned}$$

where $\cos_sim$ is cosine similarity defined as follows. Intuitively, both p and n are semantic neighbors of c ; only p is also orthographically/radically close.

For any LM with vector representation $\mathbf{e}^c$ and $\mathbf{e}^x$ for character c and x , we define cosine similarity as

$$\cos_sim(\mathbf{e}^c, \mathbf{e}^x) = \frac{\cos(\mathbf{e}^c, \mathbf{e}^x)}{\|\mathbf{e}^c\| \|\mathbf{e}^x\|}, \quad (1)$$

and, for each triplet, an **Latent Radical Score**:

$$\begin{aligned} \text{LRS}(c, p, n) \\ = \cos_sim(\mathbf{e}_m(c), \mathbf{e}_m(p)) - \cos_sim(\mathbf{e}_m(c), \mathbf{e}_m(n)). \end{aligned}$$

³We use middle-layer activations of the radicalised model as the radical embedding. This choice is supported by prior evidence, which suggests that character-level/syntactical information is most strongly represented in intermediate layers (Xu et al., 2021). Nevertheless, it is still worth investigating which layer is optimal in our specific setting.

The LRS score measures how much the model prefers the character that shares a radical with the target, when semantic similarity is held constant. If a model were organized purely by semantics, we would expect $\text{LRS}(c, p, n) \approx 0$. A positive value indicates that the model places c closer to the compositionally similar p than to an equally (or more) semantically similar but compositionally dissimilar n , suggesting an additional radical/orthographic preference. We summarize a model’s score as

$$\text{LRS}_{\text{model}} = \mathbb{E}_{(c,p,n)}[\text{LRS}(c, p, n)] \quad (2)$$

over many triplets.

3.4 Relating LRS to behavioral performance

Finally, we apply the three radical inference tasks (§3.2) to off-the-shelf Chinese LMs via prompting (using surface characters rather than radical strings) and score them with the same human evaluation protocol. We compute $\text{LRS}_{\text{model}}$ for each LM and examine correlations between LRS and task performance (generation, definition, usage). Our hypothesis is that models with stronger orthographic bias (higher LRS) will generalize better to unseen or rare characters in the radical inference tasks, providing evidence that LRS captures a meaningful notion of radical awareness.

4 Conclusion

We have outlined a psycholinguistically informed framework for probing whether Chinese language models encode and use sub-character structure. Our approach combines (i) a model trained with explicit radical decomposition as a gold standard for a radical-aware model, (ii) a structural similarity metric (LRS) that quantifies how much radical structure is encoded in standard LM embeddings beyond semantics, and (iii) controlled character inference tasks with human judgments of coarse meaning and contextual fit.

Given the limited timeline, our goal for the workshop is a focused pilot: a small but carefully constructed evaluation set, initial LRS measurements, and preliminary evidence about the extent to which contemporary Chinese LMs exhibit radical-based compositionality. We hope this work can serve as a starting point for broader benchmarks and deeper analyses of sub-character representations in Chinese and other non-alphabetic writing systems.

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